

أولاً: اختيار من متعدد (25 سؤال)

1. Where can the control layer be deployed in the cloud reference model?

- a) Only on the service layer
 - b) Only on the virtual layer
 - c) On top of the virtual layer or physical layer
 - d) Only on the physical layer
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2. The control layer primarily receives requests from:

- a) Physical layer
 - b) Security layer
 - c) Service and orchestration layers
 - d) Virtual layer
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3. Which of the following is NOT a key function of the control layer?

- a) Resource provisioning
 - b) Resource configuration
 - c) Monitoring resources
 - d) Application development
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4. Control software works in conjunction with:

- a) Service catalogs
 - b) Virtualization software
 - c) Security software
 - d) Billing software
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5. How many types of control software are defined?

- a) One
 - b) Two
 - c) Three
 - d) Four
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6. Which of the following is a type of control software?

- a) Hypervisor
 - b) Element manager
 - c) Service manager
 - d) Cloud broker
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7. Element managers are primarily used to:

- a) Manage services centrally
 - b) Manage infrastructure components independently
 - c) Perform service orchestration
 - d) Handle billing
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8. Which task can be performed by an element manager?

- a) Creating service catalogs
 - b) Installing guest OS and configuring RAID
 - c) Defining SLAs
 - d) Approving service requests
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9. Unified manager provides:

- a) Separate interfaces for each resource
 - b) A single management interface
 - c) Manual provisioning only
 - d) Billing and charging services
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10. Unified manager exposes APIs mainly to integrate with:

- a) Security layer
 - b) Physical layer
 - c) Orchestration layer
 - d) Service catalog
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11. Which phase enables unified manager to learn about available resources?

- a) Resource provisioning
- b) Resource discovery
- c) Resource grading
- d) Resource allocation

12. VLAN IDs and VSAN IDs are discovered during:

- a) Resource provisioning
 - b) Resource discovery
 - c) Pool grading
 - d) Compliance checking
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13. Resource pool grading helps in:

- a) Reducing security risks
 - b) Creating variety of services
 - c) Eliminating resource pooling
 - d) Fixing resource prices
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14. Which of the following pool grades usually has the highest cost?

- a) Bronze
 - b) Silver
 - c) Gold
 - d) All have same cost
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15. Resource provisioning starts when:

- a) Resources are discovered
 - b) Pools are graded
 - c) Consumers select services from service catalog
 - d) Element managers are installed
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16. Software-defined approach separates management functions from:

- a) Applications
 - b) Services
 - c) Infrastructure components
 - d) Users
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17. Which component abstracts underlying hardware and pools resources?

- a) Hypervisor
 - b) Software-defined controller
 - c) Element manager
 - d) Service manager
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18. A key benefit of software-defined approach is:

- a) Increased CAPEX
 - b) Reduced automation
 - c) Improved business agility
 - d) Manual resource provisioning
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19. Which resource allocation model defines quantitative bounds?

- a) Relative allocation
 - b) Absolute allocation
 - c) Dynamic allocation
 - d) Elastic allocation
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20. Lower bound in resource allocation guarantees:

- a) Maximum resource usage
 - b) Minimum resource allocation
 - c) Unlimited scalability
 - d) Resource overcommitment
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21. Hyper-threading makes a processor appear as:

- a) Multiple physical CPUs
 - b) Two logical processor cores
 - c) Separate virtual machines
 - d) Multiple hypervisors
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22. Which technique eliminates redundant copies of memory pages?

- a) Hyper-threading
- b) Memory page sharing
- c) Dynamic memory allocation
- d) Server flash-cache

23. VM load balancing across hypervisors involves:

- a) Shutting down overloaded VMs
 - b) Migrating VMs between hypervisors
 - c) Increasing physical CPUs
 - d) Manual VM placement
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24. Server flash-cache mainly improves performance for:

- a) Write-intensive workloads
 - b) Compute-intensive workloads
 - c) Read-intensive workloads
 - d) Network-intensive workloads
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25. Traffic shaping is mainly used to:

- a) Increase traffic rate
 - b) Eliminate network storms
 - c) Limit traffic rate and avoid congestion
 - d) Provide multipathing
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ثانيًا: صح / غلط (25 سؤال)

- 26. Control layer includes software tools responsible for managing cloud infrastructure.
- 27. Control layer cannot be deployed on top of the physical layer.
- 28. Control software provides a **centralized view of cloud resources**.
- 29. Element managers can be built-in or external software.
- 30. Element managers manage all infrastructure components collectively.
- 31. Unified manager allows adding or removing resources to provisioned services.
- 32. Compliance checks are performed by the unified manager.
- 33. Resource discovery provides visibility to each infrastructure resource.
- 34. All resource pool grades have the same cost.
- 35. Resource provisioning allocates resources based on service templates.
- 36. Software-defined approach enables **centralized control of infrastructure**.
- 37. Software-defined controllers expose interfaces for external applications.
- 38. Relative resource allocation defines minimum and maximum limits.
- 39. Absolute allocation is based on proportional resource sharing.
- 40. Hyper-threading improves processor utilization.
- 41. Memory page sharing allows memory over-commitment.
- 42. Dynamic memory allocation uses an agent inside the guest OS.
- 43. VM load balancing improves performance by migrating VMs.

- 44. Server flash-cache avoids network latency for storage I/O.
 - 45. Storage pool rebalancing restripes data across all drives.
 - 46. Storage space reclamation is applicable only to thick LUNs.
 - 47. Automated storage tiering moves data based on defined policies.
 - 48. Cache tiering uses SSDs as secondary cache.
 - 49. QoS prioritizes latency-sensitive traffic.
 - 50. Multipathing provides load balancing and path failover.
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(MCQ) أولاً: اختيار من متعدد

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 - 2-c
 - 3-d
 - 4-b
 - 5-b
 - 6-b
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 - 8-b
 - 9-b
 - 10-c
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 - 25-c
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ثانياً: صح / غلط

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